Nicholas Eisenberg, Ph.D.

702-354-0866 ‖ nickeisenberg@gmail.com ‖ linkedin.com/in/nick-eisenberg ‖ github.com/nickeisenberg

Experience

Senior Data Scientist *July 2022 - Present*
Nevada National Security Site

- Devloped a YOLO-based object-detection model in Pytorch that acheived a 80% MAp evaluation on a in-house thermal-image vehicle dataset. The model was trained on AWS with four GPUs by leveraging Pytorch's DataParallel.
- Designed an anomoly-detection model in Pytorch using an LSTM variationl-autoencoder to detect component faliure in a pulse-power x-ray machine. The model accurately predicted the location of a component faliure with 87%accuracy and cut machine repair times by 50%.
- Applied the U-Net architecture with Pytorch to develop a deblurring model for a pulse-powered x-ray machine. The model was used applied by the machine operators and reduced machine configuration time by 30%.
- Took initiative to develop the companies first deep-learning interface that wrapped Pytorch.
  - Introduced a standardized apporach for training, validating and testing models that ensured reproducibility and model validity.
  - Ensured that code written by all collegues could seemlessly be integrated together.
  - Supported multi-gpu training with the use of Pytorch's DataParallel.
- Sole developer of a Django web interface, which wrapped Neo4j. The interface gave non-technical staff the ability to query and update the companies Neo4j graph-database without needing to know any query syntax.

Technical Skills

Programming Languages: Python, SQL, Bash, Cypher, Lua, LaTeX

Libraries: Pytorch, Scikit-Learn, Pandas, Scipy, Sqlalchemy, Matplotlib, Plotly, Streamlit, Django

Developer Tools: Git, AWS, Unix/Linux, Vim, Tmux, Neo4j

Education

Auburn University - Ph.D. *June 2022*
Research Area: Probability (Stochastic Processes and Partial Differential Equations)
Advisor: Le Chen

University of Nevada, Las Vegas - B.S./M.S. *December 2015 / January 2018*
Major: Applied Mathematics

Publications

1: Eisenberg, N., Chen, L. Interpolating the stochastic heat and wave equations with time-independent noise: solvability and exact asymptotics. Stoch PDE: Anal Comp (2022). https://doi.org/10.1007/s40072-022-00258-6

2: Chen, L., Eisenberg, N. Invariant Measures for the Nonlinear Stochastic Heat Equation with No Drift Term. J Theor Probab (2024). https://doi.org/10.1007/s10959-023-01302-4